LACK OF DATA CONTEXT

Any data that you collect about the physical world has limited value in isolation, without appropriate contextualization that links it with other related data and entities. Consider a data stream from an IoT sensor that measures temperature. Without additional data about what location or equipment the measurements relate to, it is impossible to make any decisions based on that data.

> ***NOTE***
>
> Data contextualization refers to enriching raw data with additional relationships and information to characterize its situation, including relationships to what the data represents, the location of the data, and the type or classification.

Many organizations have historically struggled to match and integrate the multiple disparate data sources that they maintain. With multiple systems that use different identifiers for the same physical asset, customer or event, different terminologies, and different measurement scales, it becomes very difficult to analyze and derive insights to take action on.

1.5.3 Skills gap

Building or deploying a digital twin requires not only extensive technical skills across technologies, including device hardware, networking, software and data engineering, and data science but also deep knowledge of the domain and business in which you operate, in order to understand where a digital twin can bring value to the business. The importance of this combination of broad technical and domain expertise is critical to the digital twin project, delivering on the desired outcomes.

1.5.4 Build vs. buy

As with any new software system, one of the most important considerations is whether to build a custom system or purchase an off-the-shelf solution.